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Qualification: Professional Technical Trainer Level 6

Unit of competency: Competency Based Education and Training Curriculum Development

Unit Code: B7102
Duration of Unit: 30 hours

Unit Description: This unit covers the competencies required to develop a competence based curriculum . It involves applying curriculum development concepts, developing competency-based education and training curriculum and reviewing curriculum. This standard applies in the Technical and Vocational Education and Training sector

Summary of Learning Outcomes

1. Apply curriculum development concepts
2. Develop competency based education and training curriculum
3. Review CBET curriculum

EDUCATION DEPARTMENT

Course Outline

Topic	Subtopic
Introduction	Terms and concepts Rationale for the study curriculum development Purpose of the curriculum Curriculum Dimensions
Curriculum Elements	Objectives Learning experiences Organisation of learning experiences Evaluation
Curriculum Development Models	Curriculum Models ✓ Ralph Tyler ✓ CBET
Curriculum Designs	Curriculum Designs - Learner centred design - Activity based design - Core curricular design

Occupational stardard	Training needs analysis (competency gap) - Occupation analysis - Task analysis - Occupational analysis chart Packaging occupational standards
CBET curriculum	Components of a competency based curriculum Guidelines for packaging competency based curriculum Curriculum validation Curriculum evaluation and revision Curriculum maintenance
Curriculum Change and Innovation	Types of curriculum change and innovation Factors influencing curriculum change and innovation Curriculum change and innovation process Criteria for judging change and innovation

INTRODUCTION TO CBET CURRICULUM DEVELOPMENT

Curriculum development plays a crucial role in education, as it serves as a blueprint for teaching and learning. It encompasses the design, organization, and implementation of educational programs that aim to meet the needs and goals of students, educators, and society as a whole. The process of curriculum development involves a systematic approach that takes into account various factors, including educational philosophy, learning objectives, assessment methods, and instructional strategies.

Curriculum development is not a one-size-fits-all process, as it varies across different educational levels, subjects, and cultural contexts. It requires collaboration among educators, administrators, curriculum specialists, and other stakeholders to ensure that the curriculum is relevant, engaging, and aligned with educational standards. A well-designed curriculum provides a roadmap for educators, guiding them in selecting appropriate content, designing effective instructional activities, and assessing student progress.

The goals of curriculum development extend beyond the transmission of knowledge. A comprehensive curriculum aims to foster the holistic development of students, including their cognitive, social, emotional, and physical growth. It takes into account the diverse needs, interests, and abilities of learners, promoting equity and inclusion.

Curriculum development is an ongoing process that evolves with societal changes, technological advancements, and educational research. It requires regular evaluation and revision to ensure its effectiveness and responsiveness to the ever-changing educational landscape. The development of a robust curriculum empowers students with the knowledge, skills, and attitudes necessary to succeed in the modern world, preparing them for future challenges and opportunities.

In conclusion, curriculum development is a dynamic and multifaceted process that serves as the foundation of education. It shapes what students learn, how they learn, and how their learning is assessed. By embracing the principles of curriculum development, educational institutions can

create meaningful learning experiences that prepare students to become lifelong learners, critical thinkers, and active contributors to society.

TERMS AND CONCEPTS

Education

- The term education is derived from the Greek word educare meaning to nourish or to raise children.
- A process of bringing up or leading out or making manifest the inherent potentials of a pupil.
- Note that education proceeds throughout one's lifetime and the school is not the only agency that imparts education. Though the school exerts greater influence in educating the young, other social agencies for example the home/family, religious organizations, and the press/mass media supplement its work.
- Oluoch (1982) defines education as the process of acquiring and developing desirable knowledge, skills, and attitudes.
- The education process takes place in formal, non-formal, and informal settings inside and outside the school.

Curriculum

- A curriculum is a set of courses and contents offered in an educational institution. A syllabus is a descriptive list of subjects to be taught in a class. The curriculum contains course content, objectives, methodologies, etc. The syllabus contains a list of topics to cover, details about assignments, assessments.
- Kerr (1968) 'learning which is planned and guided by the school, whether carried out in groups or individually, inside or outside the school;
- Oluoch (1982) 'all that which is planned to enable students to develop the desired knowledge skills and attitudes.
- Shiundu and Omulando (1992) 'class and out-of-class activities are carried out under the aegis of a school in response to societal demands.
- Bishop (1985) 'sum total of all experiences a pupil undergoes within or without the school'.

Education and curriculum

- Education and curriculum are related concepts.
- On the one hand education is the process of imparting desirable knowledge, skills, attitudes, and values.
- On the other hand, curriculum spells out what knowledge, skills, attitudes, and values are to be imparted, how to impart them, and how to determine the level of acquisition and development of the same.

CURRICULUM DEVELOPMENT

- Shiundu and Omulando (1992) define curriculum development 'as the planning of learning opportunities intended to bring about certain changes in the learners.
- All processes which determine how the making of curriculum proceeds from the conceptualization stage to evaluation and maintenance.
- It involves a lot of value judgments and decision-making.
- NOTE: The curriculum development process shall be discussed later in detail.

Purpose of the School Curriculum

- Translates the national goals into concrete and observable behavioral changes.
- Regulate teachers' roles, responsibilities, and activities.
- Specifies objectives that schools should attain.
- Specifies activities that take place in an institution.
- Ensures uniformity in coverage of learning activities.
- Regulates evaluation procedures of an education system.

Rationale of Curriculum development

Understand the procedures and strategies of curriculum development.

Participate effectively in the various stages of the curriculum development process…in KICD, TVET-CDACC, KNEC, etc.

Translate national goals of education into instructional objectives

Interpret curriculum programs correctly.

Curriculum development is a professional activity and therefore a responsibility of all teachers.

Through studying this course unit, you, the teacher trainee, becomes acquainted with the processes involved in developing curriculum in the various subjects and would therefore be better prepared for effective teaching.

Issues regarding teaching and the teaching profession are highlighted, given that teachers are the main implementers of the curriculum.

Curriculum dimensions

Learning outcomes

- Identify the characteristics of formal, non-formal, informal, and emerging curriculum dimensions.
- Discuss the roles of the different curriculum dimensions
- Differentiate between the curriculum dimensions.

Dimensions of the school curriculum

Dimensions are aspects or facets of a curriculum. A dimension of learning is a comprehensive model that uses what researchers and theorists know about learning to define the learning process. Its premise is that this type of thinking, what we call the dimensions of learning, are essential to successful learning.

There are four dimensions of the school curriculum:

- Formal
- Non-formal

- Informal
- Emerging

Formal dimensions

It refers to the courses of study (syllabi) which are offered in a school. It is the subject matter or learning activities structured and graded for any education system running from primary through university e.g. history, geography, Kiswahili etc. The learning activities may be carried in or outside the classroom work. They show objectives to be achieved and methods of achievement.

Functions of the formal dimension

- Provides learners with planned and desired knowledge, skills, and attitudes
- Provides a scheme for the evaluation of learners
- Helps learners to progress through identified grades
- Helps in certification
- Provides feedback on learner progress

Nonformal

Refers to organized and planned out-of-class learning activities in which the students engage in or out of the school compound e.g. games, sports, drama, clubs, and other non-academic learning experiences. Just like the formal dimension, it embodies the objectives, learning activities, and student assessment method in the education program made available to all those concerned.

Functions of the non -formal dimension

Helps students to develop good leadership skills since they are given the opportunity to organize clubs and societies

Assists in physical development especially from games and sports

Enhances development of social interaction skills especially when students from different schools.

Motivates learners to be in school because of activities based on their interests and abilities.

In formal

Refers to the created environment in which students learn unconsciously through informal learning activities that go on in the school all the time.

It involves the day-to-day experiences and educative influences on planned aspects of the environment e.g. school rules, staff, work, play, etc.

Presentation of good models in school helps children to form positive favorable behavior through imitation.

Functions of the informal dimension

Provides role models for learners

Supports the formal dimension especially in acquiring knowledge, skills, and attitudes from peers, mass media, and other unplanned sources

Provides an environment to guide and counsel learners

Emerging

These refer to the unplanned learning opportunities which occur in a school e.g. Parents' Day, speech day, prize giving, graduation day, etc.

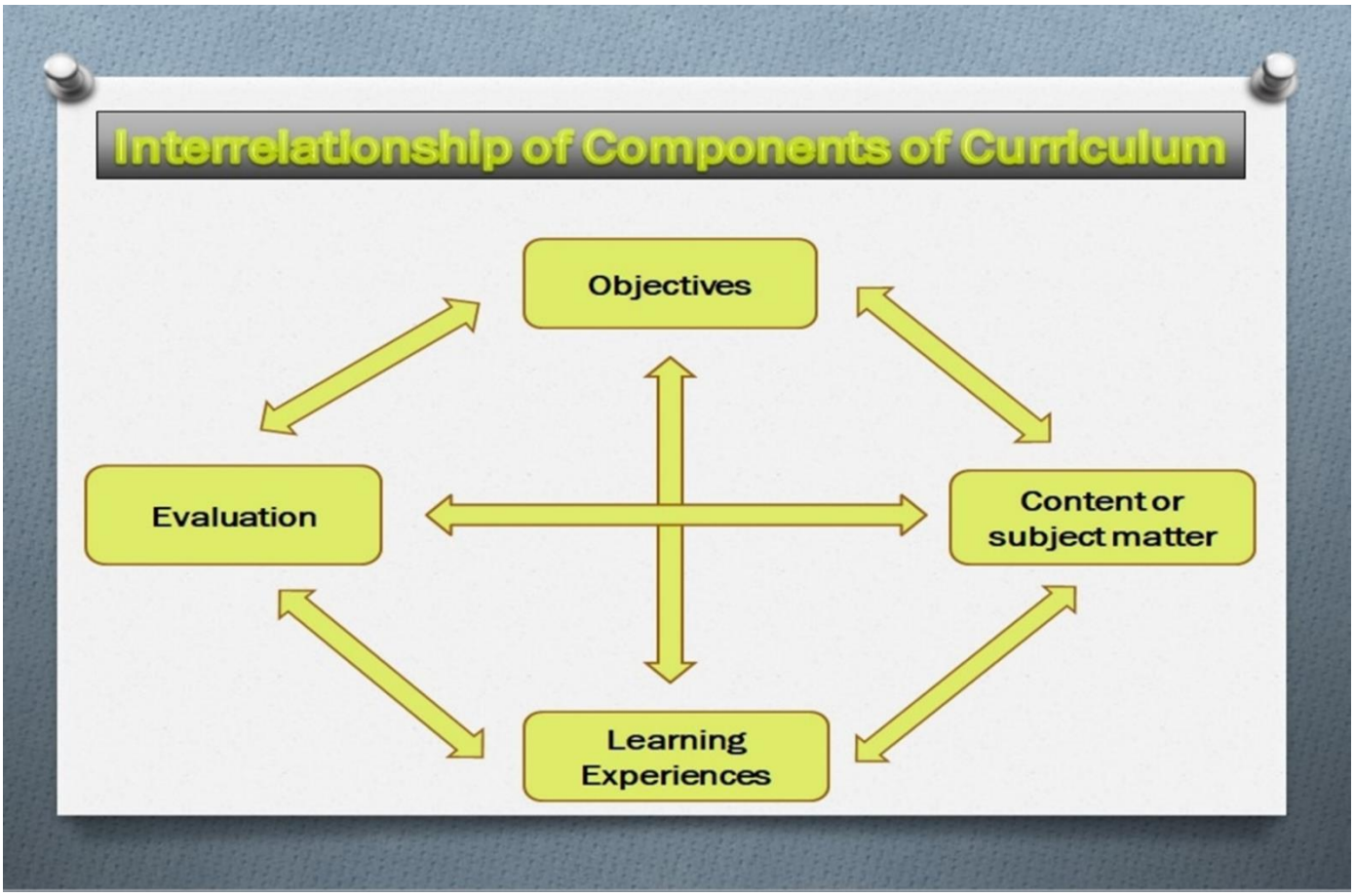
CURRICULUM ELEMENTS

This topic will take you through the curriculum elements. Specifically, you will look at the definition of curriculum elements, Curriculum objectives, Levels of objectives, Sources of objectives, Importance of objectives, and Screens of curriculum objectives. You will still continue with the curriculum elements and go-ahead to explore the Content and Learning experiences, where you will also look at Definitions, Criteria for their selection, and Criteria for their organization. The last content will be Evaluation / Assessment, where you will look at the Definition, Importance of evaluation, Types of evaluation, and Criteria of evaluation.

ELEMENTS OF A SCHOOL CURRICULUM

The curriculum is made up of four elements which are also referred to as the components of the curriculum

a. objectives
b. content
c. learning experiences
d. evaluation



These four basic elements of curriculum are essential and interrelated to each other. Aims, goals, and objectives can be simplified as “what is to be done”; the subject matter/content is “what subject matter is to be included”; the learning experiences is “what instructional strategies, resources and activities will be employed”; while curriculum evaluation is “what methods and instruments will be used to assess the results of the curriculum”.

Curriculum elements are the components that form the structure of a curriculum. They include objectives, goals, assessments, activities, materials, and evaluation. Objectives are statements that describe what students should be able to do after completing a course or unit. Goals are broad statements that define the purpose of a course or unit. Assessments are activities used to measure student understanding and progress. Activities are tasks and experiences that help students engage with the content and apply their knowledge. Materials are the resources used in a course or unit, such as lecture notes, readings, and multimedia. Evaluation is the process of assessing student learning and effectiveness of the curriculum.

CURRICULUM OBJECTIVES

Educational objectives are statements of intent describing a proposed change in the behavior of learners towards a desired direction.

When broadly stated about the education of a country they are referred to as educational goals

When they are SMART, they are instructional objectives

The Kenya National goals of education form a basis for the formulation of objectives for the various educational levels.

Curriculum objectives are expressed in three categories:

- Aims of Education
- Goals of Education
- Instructional Objectives

Aims of education

Aims are general and broad statements of intent that provide direction or intent to educational action. Educational aims guide the education system in the process of planning education content. Aims are usually written in amorphous terms using words like: learn, know, understand, appreciate, and these are not directly measurable.

Goals of education

Goals are broad statements of educational intention which are specified for each level of schooling from pre-primary education to higher education levels. They are time-bound and are stated as expected end-result of schooling at each level of Kenyan education. They may be in either amorphous language or in more specific behavioral terms.

Instructional objectives

Instructional objectives are specific, measurable, short-term, observable student behaviors. They indicate the desirable knowledge, skills, or attitudes to be gained. An instructional objective is the focal point of a lesson plan.

Developing instructional objectives

Before framing the objectives, you have to analyze;

1. The student

Knowledge level

understanding of the subject

Level of competency –

2. The content

– Level of content

– Gap analysis

Functions of Curriculum Objectives

- Define the direction of educational development
- Determine the number of levels for the program
- Inform students of the standards and expectations of the course
- Serve as an implicit contract between instructor and students, setting up a basis for accountability.
- Drive the curriculum planning
- Guide decisions about the selection of content, learning experiences, and learning strategies
- Serve as a guide in evaluating instructional progress
- Reflect the needs and aspirations of society

Sources of Curriculum Objectives

Societal needs

Analysis of the society is necessary before writing educational objectives for the following reasons:

- to enhance provision of relevant skills needed for improvement of the economy
- to facilitate transmission of acceptable cultural values
- to accommodate change – social & political

Learners interest

Learners are considered during formulation of objectives for the following reasons:

- The objectives should be appropriate to the leaner's interests, abilities (physical & mental)
- To encourage the learner to participate in the learning process
- To satisfy the needs of the learner

Academic specialist

Subject specialist are consulted during writing of educational objectives for the following reasons:

- They understand the depth of the subjects
- They are writers of books and other learning resources
- They prepare evaluation tools
- They are involved in curriculum development

Community Sources of Educational Objectives:

A community is a group of people living together in a given geographical area, and sharing common culture and a way of life, and acting collectively towards achieving a common goal. It is expedient that a school teaches the norms and values of the community where it is situated. In formulating curriculum objectives, community advisory groups might be set to present a proposal based on the peculiar problems and experiences of community members. These are then plowed into the curriculum objectives being specified especially curriculum planning at the community level. There are community resources available in educating a child. Such resources include input from parent, input from certain vocational groups, and suing field trips as important aids to learning. Certainly community input in formulating educational objectives cannot be overemphasized in fulfilling these general national objectives of education at the community level.

Psychological needs

Before a child can learn appropriately, he must adequately develop physically and mentally. The complex nature of child development has implication for curriculum planning even from the selection of objectives.

Knowledge

There is an influx of knowledge and it is only fair that we look at the school disciplines or the nature of the subject matter and the types of learning that can arise from a study of the subject matter.
Meaningful objectives are formulated in light of the available knowledge because to achieve any objective there must be positive interaction between the learner and some kind of knowledge. It is necessary to avoid formulating objectives which cannot be achieved because of the absence of appropriate knowledge content.
It is also here that we need the advice of the subject specialist on the value and application of knowledge in their subject areas.

SCREENS OF CURRICULUM OBJECTIVES

To screen means to examine or to test the curriculum objectives. Curriculum objectives are screened to find out the following:

- If they are relevant
- If they are adequate
- If they are valid
- If they are within the scope
- If they are appropriate/suitable to meet the needs of the learners and society

Types of screens

Philosophical

These constitute the values and beliefs held by a particular nation. They guide the behavior of the citizens of a given country. The philosophical orientation that the education system promotes forms an integral part of the culture of the nation. The philosophical statements usually are expressed in the manifesto of the governing political party of the day. For example in Kenya during the KANU era, the philosophical values expressed were peace, love, and unity

Psychological

Psychologists are concerned with establishing patterns of human behavior to understand and predict human behavior. They also look for determinants of behavior in hereditary as well as environmental influences. Insights gained in psychology have a bearing when developing a curriculum. Psychological screen take account of classroom behavior of learners. I.e.

- Capabilities of leaners
- Maturity
- Individual aspects etc

Sociological

Sociological screen are determined by the culture of a society, which indicated norms, values and complete requirements that are cherished by society.

Importance of objectives

- Form the basis for the development of evaluation instruments
- Guide in the selection of content / learning experiences
- Help teachers to develop or select teaching aids
- Help teachers evaluate effectiveness of teaching

CURRICULUM CONTENT AND EXPERIENCES

Curriculum content refers to organized bodies of accumulated knowledge, information and artistic heritage that learners consume with the guidance of the school.

According to Tyler, a learning experience is an interaction between the learners and the external conditions in the environment to which we react. Learning experiences include the learning opportunities provided to the learners.

Examples of learning experiences/opportunities include:

- Content or subject matter
- The teacher
- Resource persons
- Fellow learners
- Resource materials
- Teaching methods

- Learning activities that enable the learners to interact with this assorted environment of the school setting.

Curriculum content Vs curriculum experiences

Curriculum experiences are:

The mental operations that learners employ in learning the content (the two are in constant interaction, one can not deal with content without a learning experience)

The fruitful learning processes applied to content

Activities that give learners an opportunity to practice the desired behavior.

Criteria for selection of content and learning experiences

- Conformity with objectives
- Validity
- Significance
- Consistent with social realities
- Provision for meeting a wide range of objectives
- Fits the needs of the learners.

Organization of curriculum content and experiences

Organization of content and experiences determine:

1. how learning proceeds
2. how efficient and productive learning will be
3. the extent of achievement of objectives

The organization of the curriculum depends on:

1. the nature of knowledge
2. leraner growth and development
3. the learning process

Criteria for organization

1.Establishing sequence (vertical organization)

Putting the content into some order of succession

- simple to complex
- Whole to part
- chronological

- Specifics to generalizations

2. Integration (horizontal organization)

Interrelating subjects within a grade level e.g. Biology and chemistry,

3.Continuity

Providing for progressively demanding performance along the grade levels within a subject (in terms of depth and breadth).

Importance of content and learning experiences

- Give the learner an opportunity to practice the behavior implied in the objectives
- Helps to sustain the learner's interest
- Helps the learner to participate in the learning process
- Forms a basis for evaluation of learning

EVALUATION

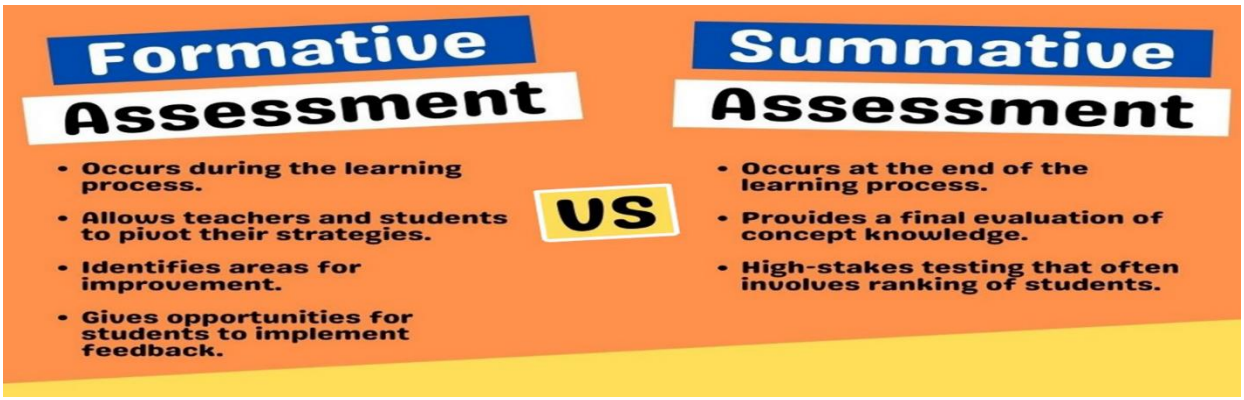
Curriculum evaluation is a method for determining the worth and effectiveness of any newly implemented curriculum. There are several stakeholders with interest in the results of curriculum evaluation that include parents, teachers, the community, administrators, and curriculum publishers.

Evaluation is a process or cluster of processes that people perform in order to gather data that will enable them to decide whether to accept, change, or eliminate something. i.e. the curriculum in general or an educational text book in particular.

Questions that have a bearing on the curriculum can only be answered through sound evaluation. Thus:

- Are the objectives relevant and realistic?
- Is the content suitable and well organized?
- Are the instructional resources relevant and adequate?
- Does the system meet the objectives as anticipated?

Types of evaluation



Purpose of formative evaluation

- Monitoring student achievement for purposes of providing individualized instruction.
- Evaluating teaching effectiveness with a view of enhancing planning, selection of teaching techniques, resources etc.
- Motivating learners
- Encouraging good studying habits
- Providing learners' entry behavior
- Providing feed back that identifies learners' strengths and weaknesses

Purpose of summative assessment

- Assigning grades
- Certification of students' achievement
- Selection of students
- Placement of students
- Guidance and counselling
- Administrative policy decisions
- Programme assessment decisions

Assessment in Competency Based Education

The concept of assessment

Educational assessment is the process of determining the extent to which learners have acquired specified knowledge, skills, values, attitudes, abilities, and competencies.

A process of establishing the extent to which the desired learning outcomes have been developed.

Guiding Principles of Competency Based Assessment

- Validity - The assessment score must reflect the learner's actual ability in the assessed criteria
- Reliability - The assessment scores should be consistent and interpreted in the same way by the assessors
- Fairness - Ensuring that the assessment process does not disadvantage any learner taking cognisance of gender, faith, cultural and socio-economic factors.
- Flexibility - Ensuring the skills, knowledge, abilities and values can be demonstrated in a variety of ways that are suitable to the school.
- Access – Providing an enabling assessment environment for learners with disabilities by making arrangements for them to demonstrate their competency levels.

Strengths of Competency-Based Curriculum

Learner-focused - Responsive and relevant to learners and enables flexibility for teachers

Focus on Competencies - More focus on competencies and less on content. Emphasises on application of knowledge, and not necessarily just its acquisition.

Flexibility- Allows greater flexibility at the local level. Teachers have autonomy in implementing the curriculum while teaching

Balance Between Formative and Summative Assessment – Allows for a range of assessment that focuses on the development of student learning outcomes, cross-curricular competencies, and literacy and numeracy

Digitally Based – Has a collaborative digital application that enables it to be improved continuously and supports learning with flexible timing and pacing through a range of learning environments.

CURRICULUM DESIGNS

Curriculum design involves the structure, organization, and arrangement of curriculum elements. It addresses several aspects to ensure a well-rounded and effective curriculum. Let's delve into each aspect:

- Scope: Scope refers to the breadth and depth of the curriculum. It involves determining the content areas, subjects, and topics that will be covered. The scope should align with the educational goals and objectives, considering the knowledge and skills that students need to acquire.

- Sequence: Sequence pertains to the order in which curriculum content is presented. It involves arranging the topics and learning experiences in a logical and progressive manner. The sequence should consider prerequisites, building upon prior knowledge, and promoting a smooth flow of learning.

- Continuity: Continuity focuses on ensuring a smooth progression of learning across different grade levels or educational stages. It involves establishing connections and building on previously acquired knowledge and skills. Continuity ensures that students can build a strong foundation and develop a coherent understanding of the subject matter.

- Integration: Integration involves making connections between different subject areas or disciplines within the curriculum. It aims to promote interdisciplinary learning and help students understand the interrelationships between various topics. Integration encourages a holistic understanding of knowledge and its application in real-world contexts.

- Balance: Balance addresses the distribution of curriculum content across different subjects or areas. It ensures that no single subject dominates the curriculum and that appropriate emphasis is given to various disciplines. Balance helps create a well-rounded education that encompasses both core subjects and elective options.

- These aspects of curriculum design work together to create a cohesive and effective curriculum. By carefully considering the scope, sequence, continuity, integration, and balance of the curriculum, educators can design a curriculum that meets the needs of learners, promotes meaningful learning experiences, and facilitates the achievement of educational goals.

There are three basic types of curriculum designs. However, there are many modifications and/or combinations of the three basic categories:

- Core curriculum design
- Activity/ experience curriculum design
- Learner centered curriculum design

CORE CURRICULUM DESIGN.

Refers to the areas of study in the school's curriculum that are required by students. It provides common learning or general education for all students.

This type of curriculum organization was supposed to promote active learning and develop a significant relationship between life and learning. The term 'core' is used in different ways. In some cases, terms like 'common learning', 'unified studies' and 'basic education' have been used to refer to the core-curriculum that is essential and is therefore a basic requirement for all students. It may refer to the compulsory subjects as opposed to the electives. The core curriculum is also seen as that part of the curriculum which is concerned with the types of experiences thought to be

necessary for effective living in society. It is the curriculum design that uses the problem-solving approach, by having the social and personal problems that are significant to the learner become part of its content. In selecting the core subjects, curriculum developers pay attention to the national philosophy and goals of education. This is why only certain subjects are given prominence and therefore become compulsory while others are electives. This is seen as an essentialist approach to curriculum development where some subjects are seen as important in the learning of others.

Relevance of the core curriculum design to education

- It helps to address the individual problems of the learner. For instance, Social Education and Ethics was meant to focus on the learners problems.
- It provides for the acquisition of basic knowledge.
- It provides for a variety of activities.
- It combines subject matter with the realities and challenges of the day.
- It is socially rather than cognitively oriented.
- It uses broader units, more flexible and freer instructional procedures and a greater variety of learning experiences than are possible in the subject-centred curriculum.
- It is possible to make a connection with the motivation force because students' experiences are largely used.

Advantages:

- Common Foundation: Provides a common foundation of knowledge and skills for all students.
- Ensures Essential Learning: Guarantees that students acquire essential knowledge and skills in key subjects.
- Scaffolded Learning: Offers a structured and scaffolded learning experience.
- Facilitates Assessment: Facilitates consistent assessment of student performance.
- Promotes Interdisciplinary Connections: Encourages interdisciplinary connections and holistic understanding.
- Streamlines Program Development: Streamlines program development and resource allocation.

Disadvantages:

- Lack of Individualization: May not cater to individual learning styles and preferences.
- Limits Specialization: Limits opportunities for in-depth specialization.
- May Become Outdated: Core curricula may become outdated over time.
- Resistant to Change: Can be resistant to incorporating new pedagogical methods or emerging topics.
- Homogeneity Concerns: This may contribute to educational homogeneity.

- Potential for Overemphasis: There is a risk of overemphasizing certain subjects or perspectives.

ACTIVITY / EXPERIENCE DESIGN CURRICULUM

Activity based learning is a wide term that refers to several models of instructions that focus the responsibility of learning to learners.

When curriculum material is translated in terms of activity, its known as activity curriculum.

The curriculum is organised around the needs and interests of the learner

The teacher has to discover the interests of the learners and help them emphasize on the most important ones

Pre-planning is not possible and considered undesirable

Cooperative planning is essential

Learners encouraged to use problem solving methods

The design discourages rigid planning

Why use activity-based curriculum?

- It is the natural urge of a child
- Activity is used as a media or means of imparting knowledge and skills.
- It is a play way of learning.
- Children are active rather than passive recipients of information.
- Activity is the greatest motivation provided to the child as they also enjoy the freedom of expressing themselves fully.
- Its core premises include the requirement that learning should be based on doing some hands-on experiments and activity.

ADVANTAGES OF ACTIVITY BASED CURRICULUM

- Student motivation is high
- Students are self-driven
- Provides for individual differences
- Students acquire necessary problem-solving skills
- It makes teaching fun.
- It helps the students remember the content better
- It makes the students feel a sense of accomplishment when the tasks are completed.
- Students can transfer the experience easily to other learning situations
- Students who are involved in activities are empowered in their learning process.

DISADVANTAGES OF ACTIVITY-BASED CURRICULUM

- Does not provide adequate preparation for life
- Lacks horizontal structure
- Lacks continuity or sequence
- Needs and interests of learners are not long-lasting and may not be valid
- Time consuming
- Schools are not well equipped.
- Its time consuming.

LEARNER-CENTRED CURRICULUM DESIGN

Learner-centered curriculum design is an educational approach that focuses on the needs, interests, and abilities of individual learners.

It places the learner at the center of the learning process, emphasizing their active engagement, motivation, and personal development.

This design of the curriculum is tailored to meet the diverse learning styles, preferences, and strengths of learners, promoting their autonomy and ownership of the learning experience.

ADVANTAGES OF LEARNER CENTERED CURRICULUM DESIGN

Individualized Learning: It emphasizes on tailoring instruction to the unique needs, interests, and abilities of each student and recognizes that learners have different learning styles, preferences, and paces of learning

Active Engagement: It encourages active engagement and participation from students and shifts the role of the teacher to a facilitator or guide who supports students in constructing their own knowledge

Relevance and Meaningfulness: It places a strong emphasis on making learning relevant to students' lives. This helps the learner to apply the knowledge and skill in the world today.

Critical thinking: Students are encouraged to analyze, evaluate, synthesize, and apply knowledge, rather than just memorize facts. This prepares them for complex problem-solving and critical thinking tasks.

Student Ownership and Autonomy: It empowers students by giving them more control and autonomy over their learning by involving them in setting goals, making decisions, and taking responsibility for their own learning thus enhancing students' self-confidence and self-motivation.

Collaborative Skills: This design incorporates collaborative learning activities, where students work together in groups or teams. This enhances communication, teamwork, and social skills and helps in understanding others

Lifelong Learning Skills: By emphasizing self-directed learning, critical thinking, and problem-solving, learner-centered curriculum design equips students with the skills and dispositions

necessary for lifelong learning. Students become active seekers of knowledge, capable of adapting to new challenges, and continuously learning throughout their lives.

DISADVANTAGES OF LEARNER CENTERED CURRICULUM DESIGN

Time-consuming: Designing a learner-centered curriculum can be time-consuming, as it requires careful consideration of individual student needs, interests, and learning styles. It may involve developing personalized learning plans and implementing differentiated instructional strategies, which can be resource-intensive for teachers.

2. Lack of coverage: In a learner-centered approach, there is a greater emphasis on individual student interests and needs. While this can enhance engagement and motivation, it may result in less coverage of standardized curriculum content. Some students may miss out on important concepts or skills if they are not explicitly included in their personalized learning plans.

3. Unequal preparation for standardized tests: Learner-centered curriculum design may not align perfectly with standardized tests or assessments, which often have a fixed content and format. As a result, students who have primarily experienced learner-centered approaches may not be adequately prepared for such assessments, potentially affecting their performance and future opportunities.

4. Limited structure and guidance: In a learner-centered curriculum, students are encouraged to take more ownership of their learning. However, some students may struggle with the autonomy and freedom provided, as they may require more structure, guidance, or explicit instructions to effectively navigate their learning journey. Without sufficient support, they may feel overwhelmed.

5. Teacher expertise and training: Implementing a learner-centered curriculum requires teachers to have a deep understanding of various instructional strategies, assessment techniques, and differentiation approaches. It may demand additional professional development and training for teachers to effectively implement and manage a learner-centered approach in the classroom.

6. Potential for unequal attention: In a classroom with a diverse group of students, a learner-centred approach may inadvertently result in unequal attention and support. Students who are more vocal, assertive, or self-directed may receive more personalized guidance and resources while those who are quieter or less confident may struggle to have their needs adequately addressed.

CURRICULUM DEVELOPMENT MODELS

Models are miniature representations that summarize data and or phenomena. They act as an aid to understanding a given process. They represent a portion of theory and therefore help in theory building.

Curriculum models have generally been limited to conceptual and graphic types.

The types of models include:

1. Ralph Tyler curriculum development model

2. CBET model (TVET CDACC)

RALPH TYLER CURRICULUM DEVELOPMENT MODEL

Also referred to as the objective's model. Developed by Ralph Tyler (1949) and published in the book "Basic Principles of Curriculum and Instruction"

Tyler raised four fundamental questions which form a basis for curriculum planning.

1. What educational purposes should a school seek to attain?
2. What educational experiences can be provided that are likely to attain this purposes?
3. How can these educational experiences be effectively organised?
4. How can we determine whether these purposes are being attained?

Tyler's questions may be reformulated into a simple 4 step process by which a curriculum is planned and developed

- Stating objectives
- Selecting experiences
- Organising experiences
- Evaluation

Note:

Ralph W. Tyler's model is one of the most influential models proposed for planning and developing curriculum. It has been referred to as a classical model because it is believed that other models drew inspiration from it.

Strengths of Tyler's Model

- It has sequential steps on the basis of time and other operations involved in the processes
- Uses student-centred approach to learning
- Designed to measure the degree to which pre-defined objectives and goals have been attained

Criticism of Tyler's model

Evaluation is fixed at the end.

Leaving evaluation till the last stage is not good enough (like doing military intelligence when the war is over). Evaluation should take place at every stage (Bruner)

The curriculum development process is seen to be linear and static.

The whole process of curriculum development should not be static but dynamic and continuously evolving (Kerr)

Difficult to harmonize objectives from different sources (learners, subject specialists and contemporary society)

Objectives appear to be fixed.

There must be a continuum of ends (objectives) and means because as we proceed, factors arise which reshape our ends in view. So, ends in view are not fixed targets but suggestions or guides (Dewey)

Dewey argued for flexibility, and freedom to change ends in view as they interact with the means.

INTRODUCTION TO COMPETENCE BASED CURRICULUM

What is a 'skill' as it applies to CBT?

Skill – ability to perform a given task to the standard of performance expected in the workplace.

What is 'knowledge' as it applies to CBT?

Knowledge is often referred to as 'underpinning knowledge'. It is the level of knowledge required to understand 'why' a skill must be performed in a specific way, how it relates to other skills and to different and more complex situations.

What is a competence?

- Ability to apply knowledge and skills to achieve intended results
- Is associated with clear ability to successfully carry out occupational activities.
- It is described in terms of 'skills', 'knowledge', and 'attitudes', as well as the typical 'context' and 'level' that a person that possesses such competence could work in.

What is competency?

A competency is defined in terms of what a person is required to do (performance), under what conditions it is to be done (conditions), and how well it is to be done (standards). Competency comprises of four dimensions:

- **Task skills:** These are the skills needed to do the task or job to the required standard
- Task management skills: This is the ability to effectively manage a number of inter related tasks all which make up the job outcome being assessed
- **Contingency management skills**: the ability to cope with irregularities and breakdowns in routine within a job or workplace
- **Job/role environment skills (outcomes)** The ability to deal effectively with the expectations of their work environment, working well with others and under instruction as required

When is a person Competent?

A person is competent when they are able to consistently apply their **knowledge** and **skills** to the standard of performance required in the workplace.

Competence-Based Education and Training (CBET)

- An organizational structure for learning/teaching that requires description in advance of the knowledge, skills, and attitudes that a student must possess upon exit from a program or course.
- Competency-based curricula clearly identify objectives, organize instruction based upon performance standards, and evaluate student performance based upon mastery of competencies.
- CBET is an approach to technical education and training that links tasks and responsibilities that professionals encounter in actual work situations to the required learning outcomes for that specific occupation.
- What a professional has to do in his work and the competences that are required, are leading elements in the curriculum development process and in the planning of the programme

CBET implies three Phases:

1. Analysis of tasks and formulation of competences needed to perform these tasks (occupational profile);
2. Development of curriculum based upon the occupational competences;
3. Facilitation (teaching) of the acquisition (learning) of these competences.

CBET Principles

- Actual workplace is leading, henceforth what to DO, why and how, is guiding principle;
- Learning outcomes relate to real tasks;
- Knowledge, skills and attitudes are integrated;
- Students are given major responsibility in their own learning
- Independent Learning

Differences between Competence Based Education versus Knowledge Based Education

Paradigm Shift

- From knowledge based to competence based;
- From teacher centered to student centered;
- From a focus on teaching to fous on learning;
- From presentations to participation in active T&L methods.

Knowledge Based Learning

- Learning outcomes derived from academic disciplines and theory;
- Knowledge, skills, attitudes seperate;
- Emphasis on theory, application thereafter;
- Teacher- centered: transferring content;
- Didactics:based on instruction, transmission;

Competence Based Learning

- Learning outcomes related to workplace tasks;
- Integrating knowledge, skills, attitudes as much as possible;
- Learner centered, teacher is facilitating learning process;
- Didactics: active learning by students;

Analysis of competencies – clarification / addition

- Always use an <u>active verb</u> to describe knowledge and skills.
- Describe the knowledge, skills or attitudes that are <u>essential</u> for completing the professional task <u>succesfully</u>

What are the characteristics of Competency Based Training (CBT)?

- Training is based on competency standards
- Learning is modular in structure
- Training delivery is individualised and self-paced

- Training is based on work that must be performed
- Training resources are directly related to the competency standards
- Assessment of learners is based on collection of evidence of the performance of work to the industry standards
- for Recognition of Prior Learning (RPL)
- The training is based on both 'on-the-job' and 'off-the-job' components
- CBT allows g allows for multiple entry and exit points and gives credits for units or modules completed

What are the advantages of CBT?

- Focuses on the 'success' of the learner
- Only deals with 'essential' content
- Learner is in control of own learning
- Continuous interaction and feedback must take place
- Learning is carried out in a sequential approach

What are the disadvantages of CBT?

- Requires significant learning resources
- The learning environment should simulate the workplace
- Critics complain that it does not suit 'higher-order' learning

What is the role of the 'learner' and the 'teacher' in CBT?

- The learner is **'self-directed'** i.e. the learner takes the initiative in what, how, and when they will learn.
- The teacher's role is that of a **'facilitator'** i.e. the teacher prepares the learning environment; they coach, prompt and help learners to reach their learning goal.

Implementation of CBET

- Change to CBET not overnight, takes time, needs commitment of all;
- TTI management plays key role in initating and supporting the process;
- Staff and students need time to get used to CBET approach;

- Industry involvement crucial;
- Implementation Plan

TVET CDACC - Curriculum Development process

The Technical and Vocational Education and Training Curriculum Development, Assessment and Certification Council (TVET CDACC) is a body corporate established under the Technical and Vocational Education and Training (TVET) Act, No. 29 of 2013. The Council is mandated to undertake design and development of Curricula for the training institutions' examination, assessment and competence certification and advise the Government on matters related thereto. This is in line with Sessional Paper No. 2 of 2015 that embraces Competency Based Education and Training (CBET) system. The Council is mandated to undertake design and development of competency based curriculum for TVET. This entails:

- Coordinating the development of competency based curriculum
- Identifying and appointing Sector Skills Advisory Committees (SSACs)
- Overseeing evaluation of curriculum and support materials for Technical and Vocational Education and Training.
- Approving proposals for application for development of competency based curriculum
- Overseeing printing, publishing and dissemination of curricula and curricula support materials

CDACC CBET Curriculum Development Process

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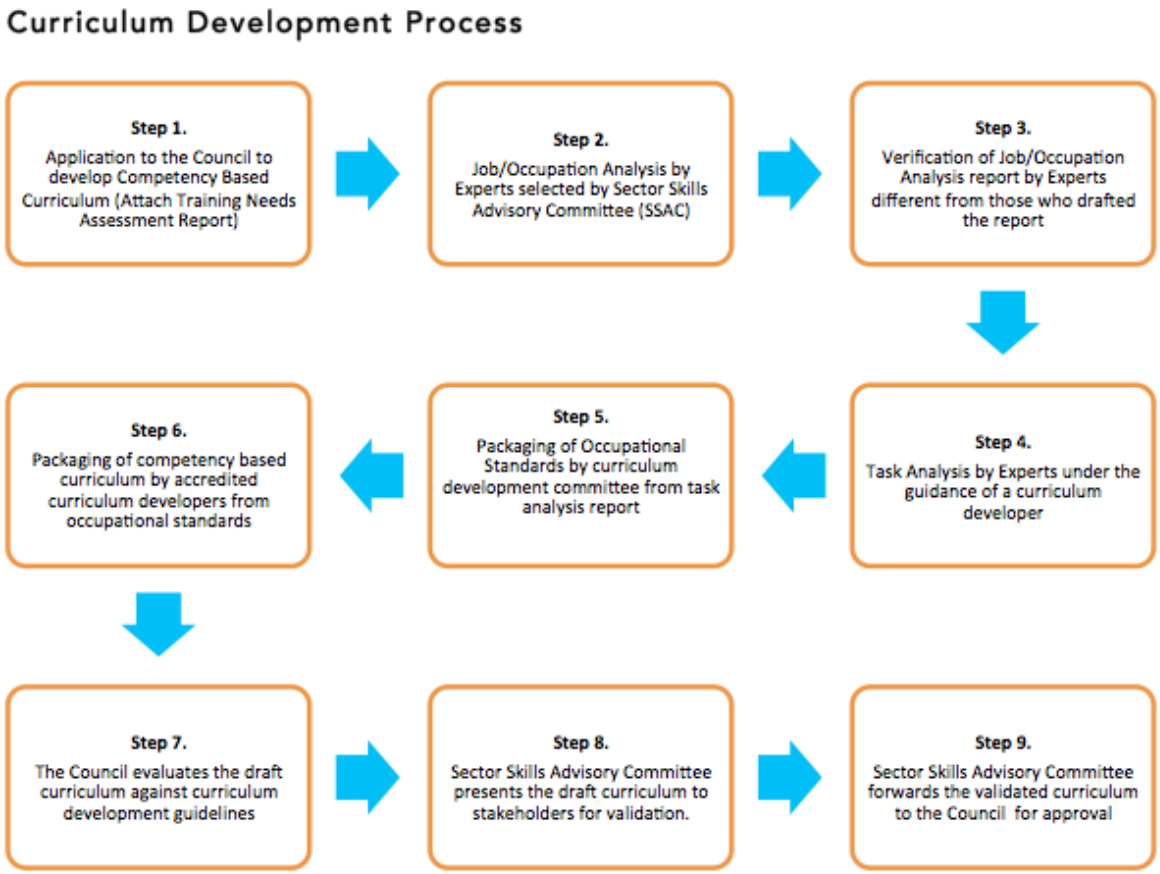
The Council has adopted the following process for development of competency based curriculum.



OCCUPATIONAL STANDARD

Occupational standards refer to a set of criteria and benchmarks that define the skills, knowledge, and competencies required for specific occupations or professions within a particular industry. These standards are developed to ensure consistency and quality in the training and education of individuals pursuing careers in technical fields. Occupational standards serve as a reference point for designing curricula, developing training programs, and assessing the proficiency of individuals in a given occupation.

Occupation refers to a person's regular work or job typically for earning a living ie Teacher, doctor Electrician, Fashion designer, cook, chef etc.

Work: tasks duties or activities that individuals perform as part of their job/occupation

Duty - General area of competence that successful workers in the occupation demonstrate or perform on an ongoing basis.

- Is the Unit of Competency in the OS and the curriculum

Task - A work activity that has a definite beginning and ending, is observable, consists of two or more definite steps, and leads to a product, service or decision.

- Is the element in the OS and learning outcome in the curriculum.

OCCUPATIONAL ANALYSIS

Occupational analysis is a systematic process of gathering, documenting, and analyzing information about a specific occupation or job.

Occupational analysis aims to provide a comprehensive understanding of the duties and tasks associated with a particular job or occupation. It involves breaking down the job into specific tasks and activities, identifying the skills and knowledge required, and considering the context and environment in which the job is performed.

DUTIES	**TASKS**					
A	**A1**	**A2**	**A3**	**A4**	**A5**	**A6**
B	B1	**B2**	**B3**	**B4**	**B5**	**B6**
C	**C1**	**C2**	**C3**	**C4**	**C5**	**C6**
D	**D1**	**D2**	**D3**	**D4**	**D5**	**D6**

A duty can have many tasks depending on the activities involved in achieving that specific duty.

How to Write a Duty or Task

Verb (Action Word) + Modifier (Adjective/Adverb, Qualifier) + Noun (Person, Place or Thing)

Examples:

a) Manage (verb) Program (modifier) Budget (noun)

b) Implement (verb) Training (modifier) Program (noun)

c) Identify (verb) garden (modifier) tools (noun)

2. Verb + Object + Condition (Context)

Example: Identify (verb) tools (object) for preparing a seedbed in a farm (context)

Example of Duty and Respective Tasks

Duty: Maintain Personal Automobile

Tasks:

- Wash car exterior
- Vacuum floor mats
- Wax car exterior
- Check fluid levels
- Check tyre air pressure

EXAMPLE OF OCCUPATIONAL ANALYSIS FOR A NURSE

Occupation	duties	Task 1	Task 2	Task 3	Task 4
Nurse	Administer Medications Safely	Administer prescribed medications accurately and document patient responses.	Evaluate the effectiveness of medications based on patient outcomes.	Adjust medication dosages under the guidance of medical practitioners.	Educate patients on the proper use and potential side effects of medications
	Communicate Effectively with Patients and Colleagues	Employ active listening skills during patient consultations.	Collaborate with interdisciplinary teams to optimize patient care.	Facilitate health education sessions for patients and their families.	Utilize technology to update and maintain accurate electronic health records

EXAMPLE 2: TECHNICAL TRAINER

OCCUPATION	DUTY	TASK 1	TASK 2	TASK 3	TASK 4	TASK 5
TECHNICAL TRAINER	CONDUCT TRAINING	Interpret curriculum	Prepare schemes of work	Prepare a lesson plan	Prepare training materials	Conduct a lesson

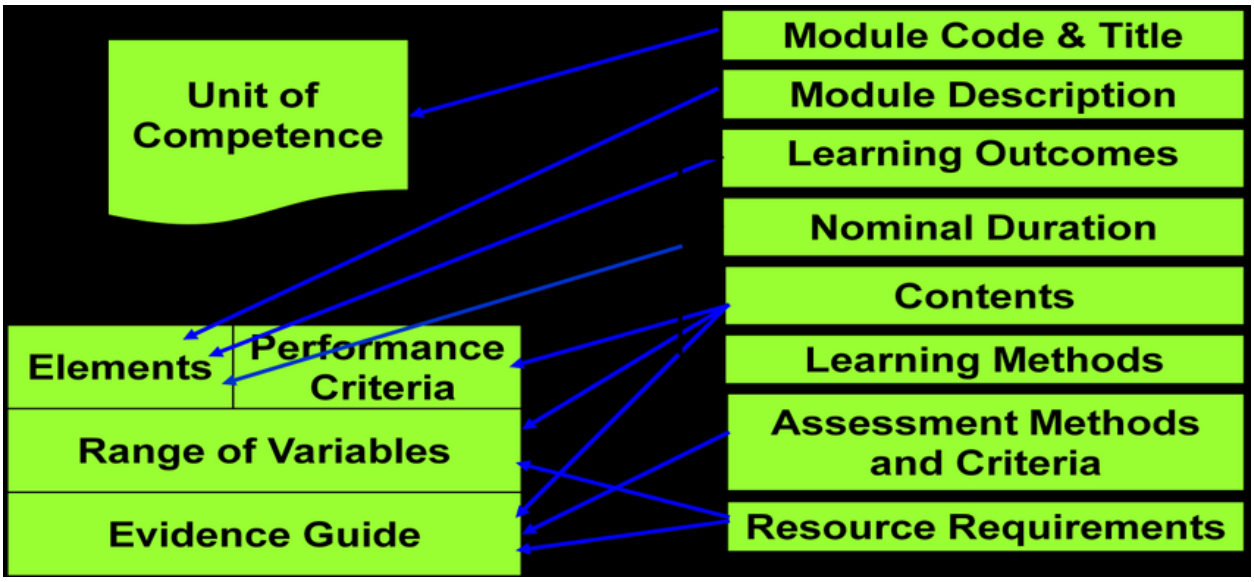
DACUM

Developing A Curriculum (DACUM) is a systematic process that analyzes a job by capturing major occupational duties and tasks that competent workers must perform, as defined by a focus group of high performing, incumbent workers, through a facilitated workshop.

OCCUPATION: TECHNICAL TRAINER								
DUTY INFORMATION		**TASK ANALYSIS (SUMMARY)**		**UNIT OF COMPETENCY INFORMATION**				
DUTIES	TASKS	ACTION	OBJECT	UNIT TITLE	ELEMENT	PERFORMANCE CRITERIA (SUMMARY)	KNOWLEDGE	SKILLS
CONDUCT TRAINING	Interpret curriculum	Interprete	Curriculum	Conduct training	Plan for a lesson	Scheme of work as per KTTC operating procedures	Demonstrate knowledge of:	Ability to: interprete curriculum
	Prepare schemes of work	Prepare	Curriculum			lesson plan as per kttc operating procedures	- curriculum interpretation	- communicate
	Prepare a lesson plan	Prepare	Lesson plan				-scheming	- prepare schemes
							-planning a lesson	-prepare a lesson plan
	Prepare training materials	prepare	Training materials, equipment and environment			Training resources as per the lesson plan and KTTC operating procedures	- assessment	- prepare training materials
							-report writing	
							-class management	-write reports
							- subject content	-assess trainees
	Conduct a lesson	Conduct	Lesson		conduct a lesson	Effective class management in		- manage class
								- carry out a

						accordance with kttc operating procedures		practical lesson -carry out a theory lesson
	Assess competencies	Assess	Competencies			Asses as per the organizational procedures		
	maintain training records	maintain	training records			Record maintenance as per the organizational procedures		

Linkage between OS and Curriculum



CURRICULUM CHANGE AND INNOVATION

Innovation is one major type of change in which something new is added to an existing curriculum. It involves initiating something new to facilitate the achievement of set goals and objectives in the educational system. Any innovation should be systematically implemented. An innovation can only be justified on the basis of achieving anticipated outcomes

FACTORS THAT NECESSITATE CURRICULUM CHANGE

In this section, we shall talk about some of the forces which affect curriculum change in a country. Some specific factors have been selected for our discussion in this topic. We shall specifically talk about social and demographic factors, economic factors, political factors, research and growth in knowledge.

i. Social and demographic factors

The nature of the population has a direct relation to curriculum planning, development and change. The society and the environment in which we live change and these changes affect the schools which are part of the society. Change in the society is also reflected in the school curriculum. Social problems such as unemployment affect people's way of life. It is the curriculum which is charged with the responsibility of ensuring that social problems are alleviated. Innovations in curriculum may come out of a need to deal with the current population problems such as the ones you have just outlined above. Public institutions, organizations and private agencies may be sources of change, or may also support change. The type and level of support will be varied, including economic support. These institutions may also resist change if it is not in favour of the well being of society. Changes in these institutions will generally affect the curriculum, either directly or indirectly.

ii. Economic factors

The existing economic order in a given country such as Kenya coupled with a new technology requires that students be introduced to new sets of skills to meet economic challenges. The state of the economy in the country will influence the curriculum. Many innovations in the school curriculum have come about due to certain economic problems. Many economic problems such as poverty and unemployment have forced educational systems to change their curricula in an attempt

to provide solutions to the problems. A curriculum which is not in line with the official economic policies may not succeed. A new curriculum requires more trained personnel, particularly teachers, more facilities, material production, text books, experiments and research to sustain the programme. This implies an increase in the national education budget. The new programme would not take off if these budgetary considerations are not put in place.

iii. Political Factors

In most countries, developed or developing, government's control and finance education. This means that they have a lot of control in curriculum development. A good relationship must exist between curriculum developers and the government in power. This is because the government can either promote or block any curriculum project which does not favour her policies and political ideology. The government provides support to curriculum changes and innovations. In circumstances where there are frequent government changes, a curriculum project or innovation could either take too long to take off or cease to exist. We should however note that curriculum projects should not contradict national policies as set out by the government.

iv. Research and growth in knowledge (Epistemology)

Research findings do affect greatly the way we teach and choose what to teach. Knowledge is an important source of curriculum content. Curriculum specialists must update themselves with new knowledge and also establish its validity and reliability. Each year there is a lot of knowledge from research. It is important to know how much of this knowledge is good enough for instructional purposes. For example, there are quite a number of theories on the nature of the child and how learning takes place. All these changes in knowledge must be considered while developing a school curriculum.

v. Technological factors

Technology has forced the school curriculum to go through change since schools are always trying to change their programmes to cope with new technology. Now look at the following questions:

- ✓ What kind of technology do you think the school can use?
- ✓ How much should the schools change in response to this technology?
- ✓ What are the implications of these changes on the functions of the school?

All these are crucial questions in planning the school curriculum. Technological changes have serious consequences for planning the curriculum. This is because some of the changes may contradict the kind of knowledge and values that society wants its youth to be given. These may also lead to changes in values and norms of society, thus putting a lot of pressure on the curriculum.

CRITERIA FOR ACCEPTING OR REJECTING AN INNOVATION

We can identify five basic criteria upon which an innovation can be judged.

i. Relative advantage

The degree to which an innovation is perceived as being better or more useful will determine its acceptability and the rate of adoption. An innovation will be considered relevant if it compares favourably or better with other practices or activities. The adopters will accept an innovation if it will improve learning, is economically manageable, has low initial cost and is less risky compared to existing standards.

ii. Compatibility

This refers to the degree to which an innovation is seen to agree with or to be supportive of the potential adopter's needs and values. It is also necessary to know how the new idea or practice fits into or supports the existing educational programme. Compatibility ensures greater security and less risk to the user, and makes the innovation more meaningful.

iii. Triability

This refers to the extent to which an innovation can be piloted, without taking too much time, energy or funds. It is better if adequate data can be obtained from a small scale trial in order to make a decision as to whether the innovation is necessary or not.

iv. Observability

Stakeholders in an innovation such as potential adopters and sponsors would like to see tangible results from their efforts. They need to know the immediate consequences of the innovation in the country. Effective accountability requires availability of concrete information.

v. Complexity

An innovation must be simple enough to be understood and utilized by the user. Complexity may lead to rejection by potential adopters. If teachers find an innovation too difficult to understand and implement, they are likely to reject it.